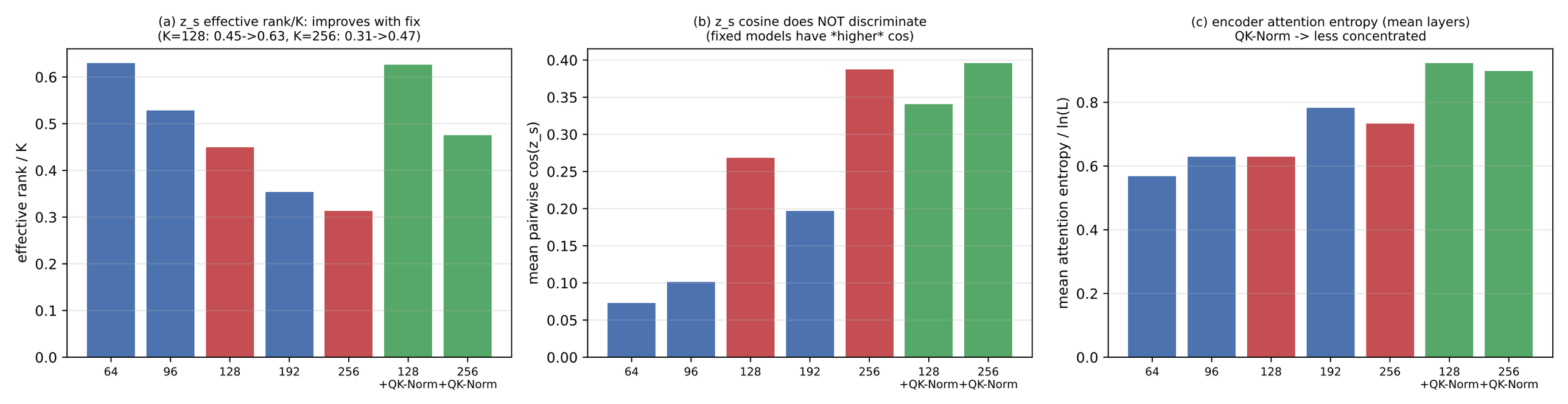
(a) z_s effective rank/K: improves with fix
(K=128: 0.45->0.63, K=256: 0.31->0.47)
effective rank / K
0.0
0.1
0.2
0.3
0.4
0.5
0.6
64
96
128
192
256
128
+QK-Norm
256
+QK-Norm
(b) z_s cosine does NOT discriminate
(fixed models have *higher* cos)
mean pairwise cos(z_s)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
64
96
128
192
256
128
+QK-Norm
256
+QK-Norm
(c) encoder attention entropy (mean layers)
QK-Norm -> less concentrated
mean attention entropy / ln(L)
0.0
0.2
0.4
0.6
0.8
64
96
128
192
256
128
+QK-Norm
256
+QK-Norm